

Alfido Tech | Sales Performance Analysis

Project summary and implementation guide

Dataset: Kaggle Superstore Sales — a sample retail transaction dataset. This document is a concise companion to the executable notebook. Dataset-specific results should be generated by running the notebook against the downloaded file; no numerical findings are asserted here without that execution.

Objectives and analytical coverage

- Prepare transaction data: normalize fields, parse dates, coerce measures, inspect missing values, remove exact duplicate rows and validate sales/quantity.
- KPIs: revenue, profit, profit margin, distinct orders, average order value and units sold.
- Performance cuts: region, category, subcategory and product; identify high-revenue and low-profit products.
- Time analysis: monthly revenue trend and calendar-month patterns across observed years.
- Discount analysis: compare observed discount levels with sales/profit descriptively; do not interpret association as causation.

KPI definitions

Metric	Definition
Revenue	Sum of line-item Sales
Profit	Sum of Profit; preserve negative values
Profit margin	Total Profit ÷ Total Sales
Average order value (AOV)	Total Sales ÷ distinct Order IDs
Conversion rate	Requires a defined conversion event and visitor/session/lead denominator; otherwise

Five tactical opportunities for Alfido Tech

- 1. Protect contribution, not just revenue.** Use product-level profit and margin to identify high-sales/low-profit items. Review landed costs, discount ceilings and pricing for those SKUs before scaling promotion.
- 2. Localize regional offers and inventory.** Compare regional revenue, profit and order mix. Align stock, fulfillment capacity and region-specific campaigns with observed demand, while testing incremental lift.
- 3. Build a calendar-led campaign plan.** Use monthly/quarterly patterns to plan inventory and campaign timing. Validate recurring peaks across years and avoid overreacting to one-off spikes.
- 4. Rationalize the assortment using product evidence.** Prioritize replenishment and merchandising for products with sustained revenue and positive profit; review persistently loss-making or low-demand items for supplier terms, bundling or discontinuation tests.
- 5. Introduce controlled discount experiments.** Segment discounts and monitor profit per order alongside conversion and basket size. Run holdout/A-B tests with guardrails rather than assuming deeper discounts create profitable growth.

Dashboard / chart set

The notebook produces KPI output, revenue by region, category and subcategory charts, top and bottom product tables, a monthly revenue line chart, calendar-month comparison, and discount-versus-profit scatter plot when the corresponding fields exist. Exported tables can be loaded into Tableau or Power BI if a native

BI dashboard is required.

Limitations and responsible interpretation

This dataset is a generic Superstore case study and should not be represented as Alfido Tech's own sales. The notebook reports actual calculations only after the file is loaded. Conversion cannot be inferred from orders alone. Seasonality should be assessed with sufficient years and comparable periods. Product rankings based on observed revenue do not automatically imply profitability or strategic importance.

Source: <https://www.kaggle.com/datasets/bhanupratapbiswas/superstore-sales>